[METRIC]

MIL-D-218B
13 May 1981
SUPERSEDING
MIL-D-218A
18 August 1960

MILITARY SPECIFICATION

DIBUTYL PHTHALATE, TECHNICAL (METRIC)

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one technical grade of dibutyl phthalate.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

FEDERAL

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies)

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

FSC 6810

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

D891 - Specific Gravity of Industrial Aromatic Hydrocarbons and Related Materials
D1193 - Reagent Water
D1209 - Color of Clear Liquids (Platinum-Cobalt Scale)
D1613 - Acidity in Volatile Solvents and Chemical Intermediates Used in Paint, Varnish, Lacquer, and Related Products
D1617 - Ester Value of Solvents and Thinners
D1951 - Ash in Drying Oils and Fatty Acids

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Chemical and physical characteristics. The dibutyl phthalate shall conform to the chemical and physical characteristics of table I when tested as specified therein.

TABLE I. Chemical and physical characteristics

Characteristic	Requirement		Test paragraph
	Minimum	Maximum	
Specific gravity, 20°/20°C	1.044	1.054	4.2.4.1
Ester content (as dibutyl phthalate), percent by weight	99		4.2.4.2
Acidity (as phthalic acid), percent by weight		0.03	4.2.4.3
Ash, percent by weight		0.02	4.2.4.4
Color, platinum-cobalt scale		20	4.2.4.5

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the dibutyl phthalate produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For test. Sampling shall be conducted in accordance with table II. A representative specimen of approximately 0.5 liter shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

TABLE II. Sampling for test

Number of containers in batch or lot	Number of sample containers
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled unit, intermediate, or shipping container, as applicable, ready for shipment. Sample unit, intermediate, and shipping containers shall be examined for the following defects using an AQL of 4.0 percent defective:

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(a) Container damaged or leaking
(b) Unitization not as specified
(c) Marking incorrect, missing, or illegible

4.2.3.2 For test. Approximately equal portions of all the specimens taken in 4.2.2.2 shall be thoroughly mixed to form a composited specimen of no less than 1.0 liter. The composited specimen shall be tested as specified in 4.2.4. Each test shall be conducted in duplicate analysis. Failure of either analysis of any test shall be cause for rejection of the lot represented.

4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Specific gravity. Determine the specific gravity of the specimen at 20°/20°C by any convenient method that is accurate to the third decimal place. Methods A, B, and C of ASTM D891 are suitable for this determination.

4.2.4.2 Ester content. Determine the percent by weight ester content of the specimen in accordance with ASTM D1617 using the reaction at 98°C procedure and titrating with 0.5N hydrochloric acid.

4.2.4.3 Acidity. Determine the percent by weight acidity of the specimen as phthalic acid in accordance with ASTM D1613 using alcohol as the solvent. Calculate the acidity as percent by weight phthalic acid from the acidity as milligrams potassium hydroxide per gram of sample as follows:

$$\text{Percent phthalic acid} = \frac{A}{6.7}$$

where: A = Acidity as milligrams potassium hydroxide per gram of sample.

4.2.4.4 Ash. Determine the percent by weight ash of the specimen in accordance with ASTM D1951.

4.2.4.5 Color. Determine the color of the specimen on the platinum-cobalt scale in accordance with ASTM D1209.

5. PACKAGING

5.1 Interplant shipment. Uniform quantities of dibutyl phthalate shall be unit packed and packed to prevent damage in shipment from the supply source to the first receiving activity for immediate use or further processing. Shipping containers shall be in accordance with the rules and regulations applicable to the mode of transportation.

5.2 Unitization. Containers shall be unitized in a manner to assure carrier acceptance, safe arrival at destination, and stability of the stacks of the unitized containers in storage.

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5.3 Marking. Containers shall be marked in accordance with Fed. Std. No. 123 and shall include the contract or purchase order number, lot or batch number, date of manufacture, and name and address of the prime contractor. Unitized loads shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The dibutyl phthalate covered by this specification is intended for use in the manufacture of smokeless powder.

6.2 Ordering data. Procurement documents should specify the title, number, and date of this specification.

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E29.

Custodians:

Army - EA
Navy - OS

Review activities:

Army - MD, MI
DLA - GS

Preparing activity:

Army - EA

Project No. 6810-0613

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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